

CIRCUIT THEORY

Circuit Variables

→ Circuit theory is a special case of electromagnetic field theory: the study of static and moving electric charges.

→ If $\lambda \gg d$ (the wavelength is much greater than the physical dimension of the circuit) then we can use the circuit theory instead of electromagnetic field theory to find currents and voltages. The circuit theory provides simple solutions (of sufficient accuracy) to problems that would become hopelessly complicated if we were to use electromagnetic field theory.

→ If $f = 50 \text{ Hz}$ $\lambda = \frac{c(\text{speed of light})}{f(\text{frequency})} = \frac{3 \times 10^8 \text{ m/s}}{50 \text{ 1/s}} = 6000 \text{ km}$

→ if $\lambda \gg d$ circuit analysis can be employed to obtain a solution otherwise we must use electromagnetic field theory (need to solve Maxwell's equations.)

→ If $\lambda > 10d$ we assume $\lambda \gg d$

Voltage and Current

→ The Concept of electric charge is the basis for describing all electrical phenomena.

→ The charge is bipolar, meaning that electrical effects are described in terms of positive and negative charges.

→ The electric charge exists in discrete quantities, which are integral multiples of the electronic charge, $1.602 \times 10^{-19} \text{ C}$.

→ Electric effects are attributed to both the separation of charge and charges in motion.

→ In circuit theory, the separation of charge creates an electric force (voltage), and motion of charge creates an electric fluid (current).

→ Voltage is the energy per unit charge created by the separation.

→ The rate of charge flow is known as the electric current.

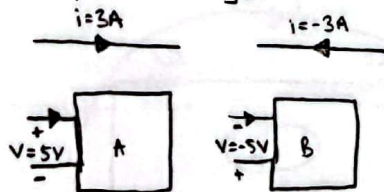
$V = \frac{dw}{dq}$ V , voltages in volts w , energy in joules
 q , charge in coulombs

$i = \frac{dq}{dt}$ i , current in amperes t , time in seconds
 q , charge in coulombs

→ The power is defined as energy per unit time.

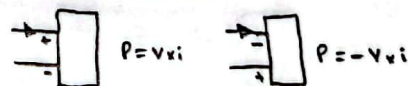
$P = \frac{dw}{dt} = \frac{dw}{dq} \frac{dq}{dt} = V \times i$ P , power in watts

→ You should define a reference direction while writing the equations or talking about the current. The same is true for the voltage.



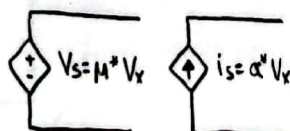
Terminal A is 5 volt positive with respect to terminal B.

→ If the current enters from positive side $P = V \times i$, otherwise $P = -V \times i$.

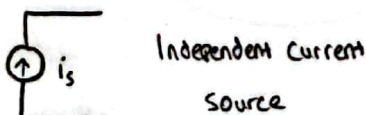


→ If $P > 0$, circuit element absorbs (dissipates) power. Otherwise it delivers (generates) power.

Voltage and Current Sources



Voltage controlled dependent sources



Current controlled dependent sources

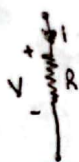
Electrical Resistance (Ohm's law)

$V = R \times i$ R , resistance in Ohms

→ Most materials exhibit measurable resistance to current.

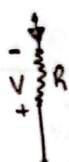
The amount of resistance depends on the material. Most metals such as copper and aluminum have small values of resistance.

→ For the purposes of circuit analysis, we must reference the current in the resistor to the terminal voltage. We can do so in two ways: either in the direction of the voltage drop across the resistor or in the direction of the voltage rise across the resistor.



$$V = R \cdot i$$

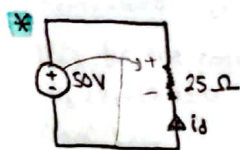
$$P = v \cdot i = i^2 R = v^2 / R$$



$$V = -R \cdot i$$

$$P = -v \cdot i = -(-R \cdot i) = i^2 R = v^2 R$$

→ The ratio of current to voltage is also a constant and written as $\frac{i}{V} = G$ is called **Conductance** and denoted by σ (inverted Ω).
The unit is mho or Siemens (S).



$$i_d = -50/25 = -2A$$

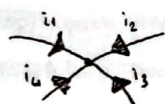
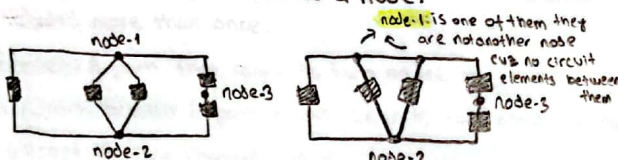
$$P = -50(-2) = 100W$$

parallel olduklan
i_d'n

Kirchoff's Current Law

→ The algebraic sum of all the currents at any node in a circuit equals zero.

→ **Node:** A point which two or more circuit elements have a common connection is called a node.



$$\sum_{k=1}^n i_k = i_1 + i_2 + i_3 + i_4 = 0$$



$$\sum_{k=1}^n i_k = i_1 - i_2 - i_3 + i_4 = 0$$

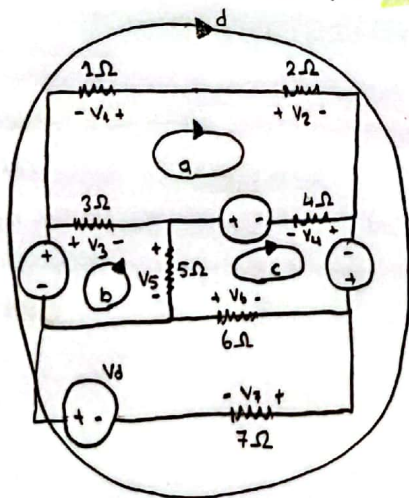
Sum of the currents entering the node

$$\sum_{k=1}^n i_k = -i_1 + i_2 + i_3 - i_4 = 0$$

Sum of the currents leaving the node

$$i_1 + i_4 = i_2 + i_3$$

→ In any circuit with n nodes, $n-1$ independent current equations can be derived from KCL.



KVL for

$$\text{Loop a: } -V_1 + V_2 + V_4 - V_6 - V_3 = 0$$

$$\text{Loop b: } -V_4 + V_3 + V_5 = 0$$

$$\text{Loop c: } V_6 - V_4 - V_6 - V_5 = 0$$

$$\text{Loop d: } -V_4 - V_1 + V_2 - V_6 + V_7 - V_8 = 0$$

Some alternative statements for the law.

→ The algebraic sum of all the currents entering a node in a circuit equals zero.

→ The algebraic sum of all the currents leaving a node in a circuit equals zero.

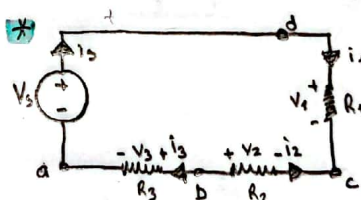
→ The sum of all the currents leaving a node in a circuit equals to sum of the currents entering the node.

Kirchoff's Voltage Law

→ The algebraic sum of voltages around any closed path in a circuit equals zero.



$$V_2 - V_1 = 0 \quad V_2 = V_1$$



Ohm's law

$$V_1 = i_1 \cdot R_1$$

$$V_2 = i_2 \cdot R_2$$

$$V_3 = i_3 \cdot R_3$$

KCL

$$\text{Node a: } i_5 = i_3$$

$$\text{Node b: } i_3 = -i_2$$

$$\text{Node c: } i_2 = -i_1$$

$$\text{Node d: } i_5 = i_1$$

$$i_5 = i_1 = -i_2 = i_3$$

KVL

$$V_1 - V_2 + V_3 - V_5 = 0$$

$$i_1 \cdot R_1 - i_2 \cdot R_2 + i_3 \cdot R_3 - V_5 = 0$$

$$i_5 \cdot R_1 + i_5 \cdot R_2 + i_5 \cdot R_3 - V_5 = 0$$

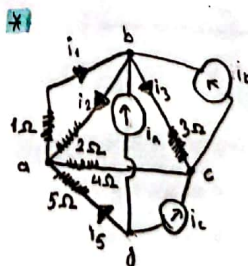
KCL for

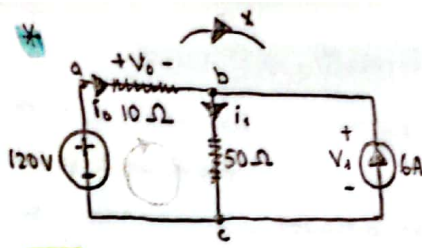
$$\text{Node A: } i_1 + i_2 + i_4 - i_5 = 0$$

$$\text{Node B: } -i_1 + i_2 + i_3 - i_4 = 0$$

$$\text{Node C: } -i_3 - i_4 + i_5 - i_6 = 0$$

$$\text{Node D: } i_5 + i_6 + i_7 = 0$$





KCL for the node b

$$i_1 - i_0 - 6 = 0 \quad i_0 = i_1 - 6$$

KVL for the node x

$$V_0 + V_1 - 120 = 0$$

Ohm's Law

$$V_0 = 10i_0 \quad V_1 = 50i_1$$

$$10(i_1 - 6) + 50i_1 = 120$$

$$i_1 = 3A \quad i_0 = -3A \quad V_1 = 150V$$

find the i_0 , power?

power dissipated in the resistors

$$P_{50\Omega} = 3^2 \cdot 50 = 450W$$

$$P_{10\Omega} = (-3)^2 \cdot 10 = 90W$$

power delivered by the sources

$$P_{120V} = 120 \cdot i_0 = 360W$$

$$P_{6A} = -V_1 \cdot 6 = -900W$$

generated $\rightarrow 900W$

dissipated $\rightarrow 450 + 90 + 360 = 900W$

CHAPTER 4

Node: A point where two or more circuit elements join.

Essential Node: A node where three or more circuit elements join.

Path: A trace of adjoining basic elements with no elements included more than once.

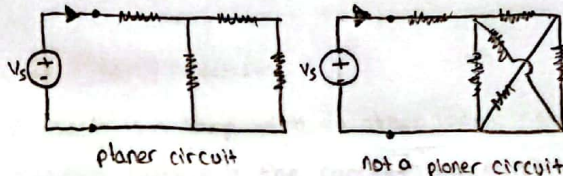
Branch: A path that connects two nodes.

Essential Branch: A path which connects two essential nodes without passing through an essential node.

Loop: A path whose ending is the same as the starting node.

Mesh: A loop that does not enclose any other loop.

Planar circuit: A circuit that can be drawn on a plane with no crossing branches.



The Systematic Approach

The number of unknown currents in a circuit equals to the number of essential branches, b_e , where current is not known.

n_e is the number of essential nodes.

We can derive $n_e - 1$ independent equations by applying KCL.

Need $b_e - (n_e - 1)$ more equations. We can write $b_e - (n_e - 1)$ equations using KVL.

Simple Resistive Circuits

Resistors in Series

$\rightarrow$ When just two circuit elements connect at a single node, they are said to be in series.

$\rightarrow$ Series-connected circuit elements carry the same current.

R_1 and R_2 connect at the b node, $i_1 = i_2$ $R_{eq} = R_1 + R_2$

Resistors in Parallel

$\rightarrow$ When two circuit elements connect at a single node pair, they are said to be in parallel.

$\rightarrow$ Two circuit elements, they are parallel to each other, share the same voltage.

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} \quad i_s = \frac{V_s}{R_{eq}} \quad G_1 = \frac{1}{R_1} \quad G_{eq} = G_1 + G_2$$

$$i_1 = i_2 = i_s \quad i_1 = \frac{V_s}{R_1} \quad i_2 = \frac{V_s}{R_2}$$

$$-V_s + i_1 R_1 + i_2 R_2 = 0 \quad i = \frac{V_s}{R_1 + R_2}$$

$$V_1 = R_1 \cdot i = V_s \left(\frac{R_2}{R_1 + R_2} \right)$$

$$V_2 = R_2 \cdot i = V_s \left(\frac{R_1}{R_1 + R_2} \right)$$

$$i_1 = i_s \cdot \frac{R_2}{R_1 + R_2} \quad i_2 = i_s \cdot \frac{R_1}{R_1 + R_2}$$

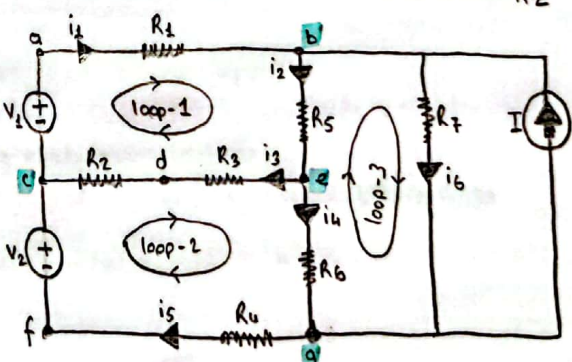
$$R_1 = \frac{R_a R_c}{R_a + R_b + R_c} \quad R_2 = \frac{R_a R_b}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_b R_c}{R_a + R_b + R_c}$$

$$R_a = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_3}$$

$$R_b = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_1}$$

$$R_c = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_2}$$



There are four essential nodes (b, c, e, g). We can write three independent equations using KCL.

for node b: $-i_1 + i_2 + i_6 - I = 0$ for node c: $i_1 - i_3 - i_5 = 0$

for node e: $i_3 + i_4 - i_2 = 0$

There are six essential branches. We need three more equations.

There are four loops. Using KVL we can write three independent eqs.

for loop-1: $R_1 \cdot i_1 + R_5 \cdot i_2 + (R_2 + R_3) i_3 - V_1 = 0$

for loop-2: $-(R_2 + R_3) i_3 + R_6 \cdot i_4 + R_4 \cdot i_5 - V_2 = 0$

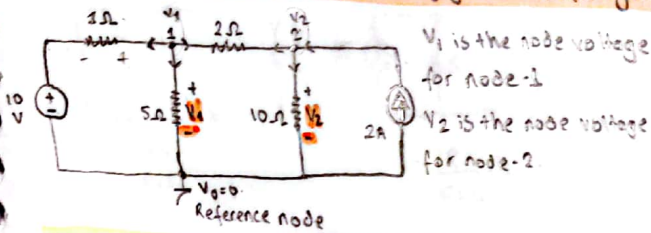
for loop-3: $-R_5 i_2 + R_7 i_6 - R_6 i_4 = 0$

Node-Voltage Method

→ This method enable us to describe a circuit in terms of $(n-1)$ equations.

→ A node voltage is defined as the voltage rise from the reference node to a nonreference node.

* Node'den bütün okunlar çıkıyor varsayıyoruz. 0'a eşitliyoruz.



KCL for node-1

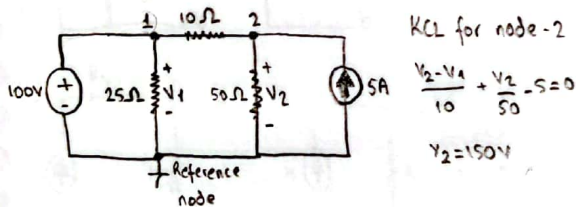
$$\frac{V_1 - 10}{1} + \frac{V_1}{5} + \frac{V_1 - V_2}{2} = 0$$

KCL for node-2

$$\frac{V_2 - V_1}{2} + \frac{V_2}{10} - 2 = 0$$

Node-Voltage: Some special cases

→ When a voltage source is the only element between two essential nodes;



Mesh-Current Method

→ This method enable us to describe a circuit in terms of $(n-1)$ equations.

→ A mesh is a loop with no other loops inside it.

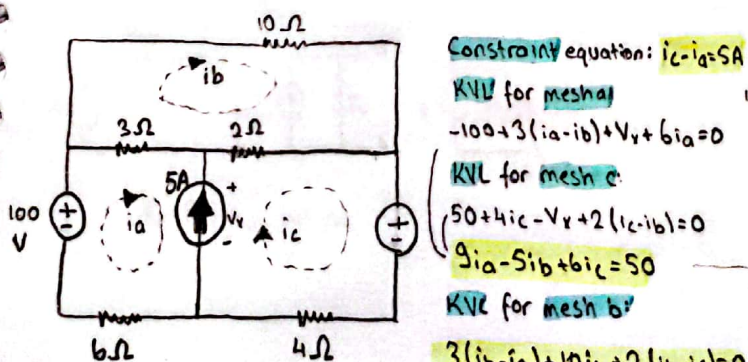
→ A mesh current is the current that exist only in the perimeter of a mesh.

→ The circuit must be planar.

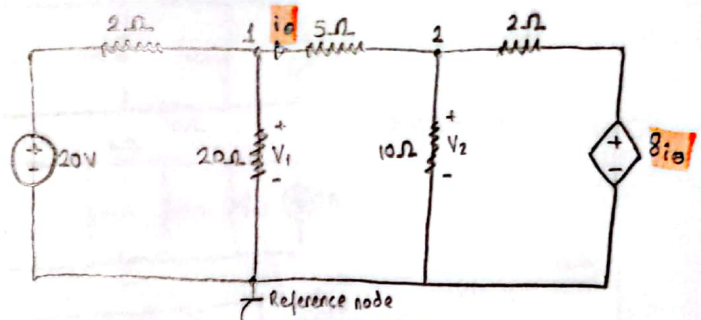


Mesh-Current Method: Some special cases

→ When a branch includes a current source, the mesh current method requires some additional manipulations.



Node-Voltage Method and Dependent Sources



Constraint equation (i_1)

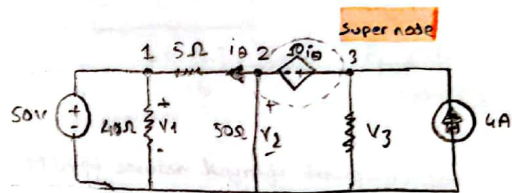
$$i_1 = \frac{V_1 - V_2}{5}$$

KCL for node-1

$$\frac{V_1}{20} + \frac{V_1 - 20}{2} + \frac{V_1 - V_2}{5} = 0$$

KCL for node-2

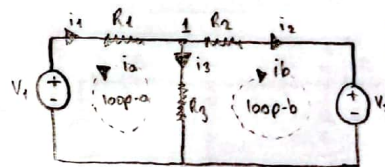
$$\frac{V_2}{10} + \frac{V_2 - 8i_1}{2} + \frac{V_2 - V_1}{5} = 0$$



we can directly write equation-1 by combining node-2 and node-3

$$\frac{V_2 - V_1}{5} + \frac{V_2}{50} + \frac{V_3}{100} - 4 = 0$$

constraint equation 1: $i_1 = \frac{V_2 - 50}{5}$
 constraint equation 2: $V_3 = V_2 + 10i_1$
 $= 3V_2 - 100$
 $V_2 = 60, V_3 = 80$



KCL for node-1

$$i_1 = i_2 + i_3$$

KVL for loop-a

$$-V_1 + i_1 \cdot R_1 + i_3 \cdot R_3 = 0 \Rightarrow V_1 = i_1 \cdot R_1 + i_3 \cdot R_3$$

KVL for loop-b

$$V_2 - i_3 \cdot R_3 + i_2 \cdot R_2 = 0 \Rightarrow -V_2 = i_2 \cdot R_2 - i_3 \cdot R_3$$

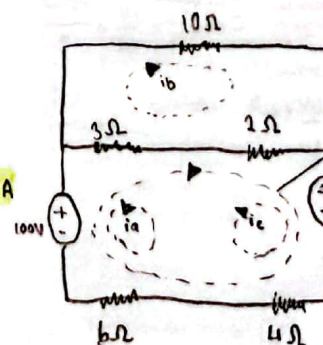
Using mesh currents for loop-a

$$-V_1 + R_1 \cdot i_1 + R_3 \cdot (i_1 - i_2) = 0 \Rightarrow V_1 = R_1 \cdot i_1 + R_3 \cdot (i_1 - i_2)$$

Using mesh currents for loop-b

$$V_2 = R_3 \cdot (i_2 - i_1) + R_2 \cdot i_2 = 0 \Rightarrow -V_2 = R_3 \cdot (i_1 - i_2) + R_2 \cdot i_2$$

$$i_1 = i_a, i_2 = i_b, i_3 = i_a - i_b$$



Supermesh

we can write equation-1 directly

KVL for Supermesh

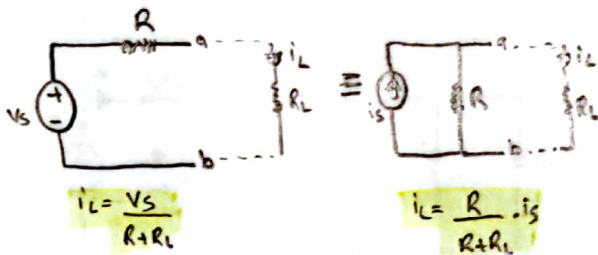
$$-100 + 3(i_1 - i_2) + 2(i_2 - i_3) + 50 + 4i_1 + 6i_1 = 0$$

$$9i_1 - 5i_2 + 6i_3 = 50$$

Source Transformation

→ A source transformation allows a voltage source in series with a resistor to be replaced by a current source in parallel with the same resistor or vice versa.

Assume R_L is connected between nodes a and b

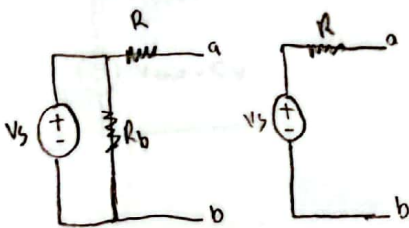


$$i_L = \frac{V_S}{R + R_L}$$

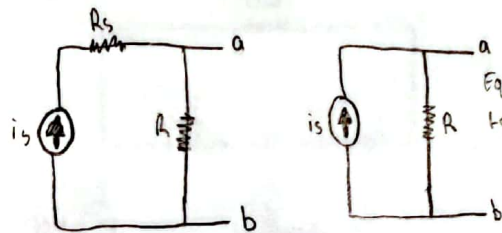
$$i_L = \frac{R}{R + R_L} \cdot I_S$$

$$\frac{V_S}{R + R_L} = \frac{R}{R + R_L} \cdot I_S$$

$$I_S = \frac{V_S}{R} \text{ or } V_S = I_S \cdot R$$



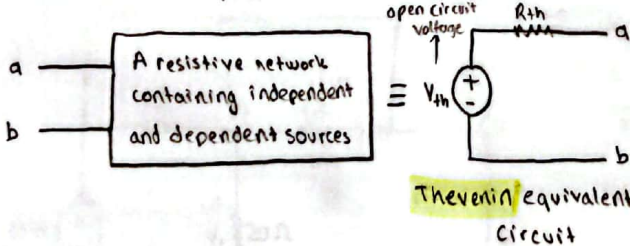
Equivalent with respect to terminals a and b.



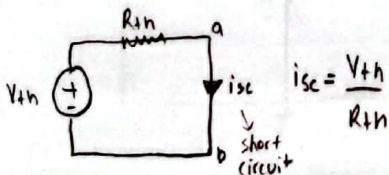
Equivalent with respect to terminals a and b.

Thevenin and Norton Equivalents

→ Thevenin and Norton equivalents are circuit simplification techniques that focus on terminal behavior.

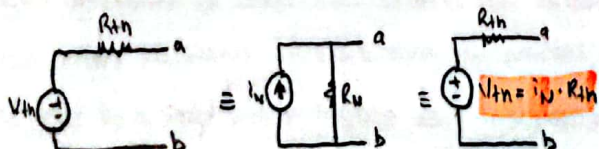


Thevenin equivalent circuit

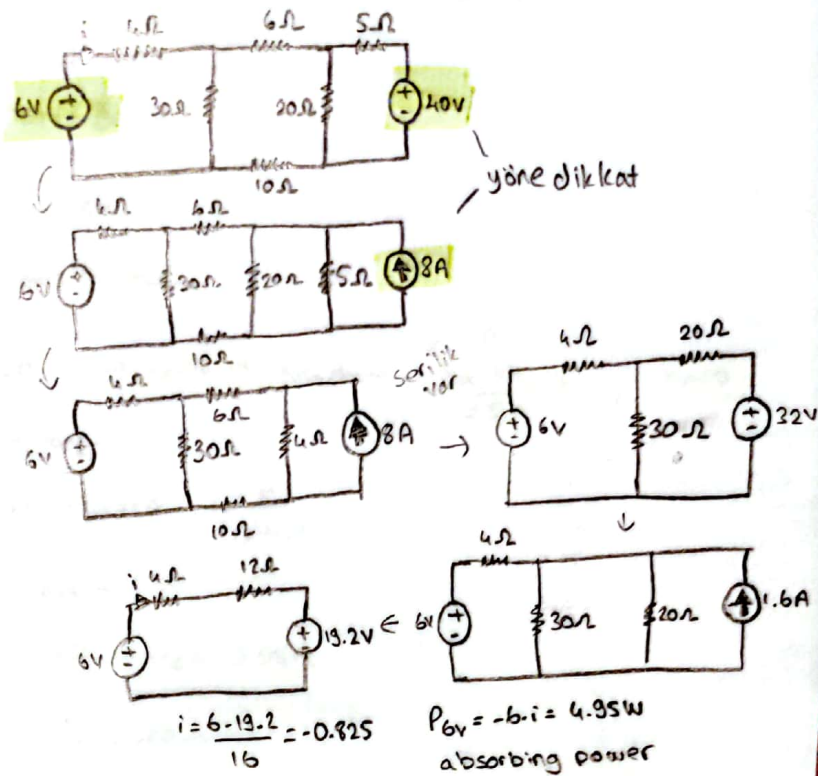


$$i_{sc} = \frac{V_{th}}{R_{th}}$$

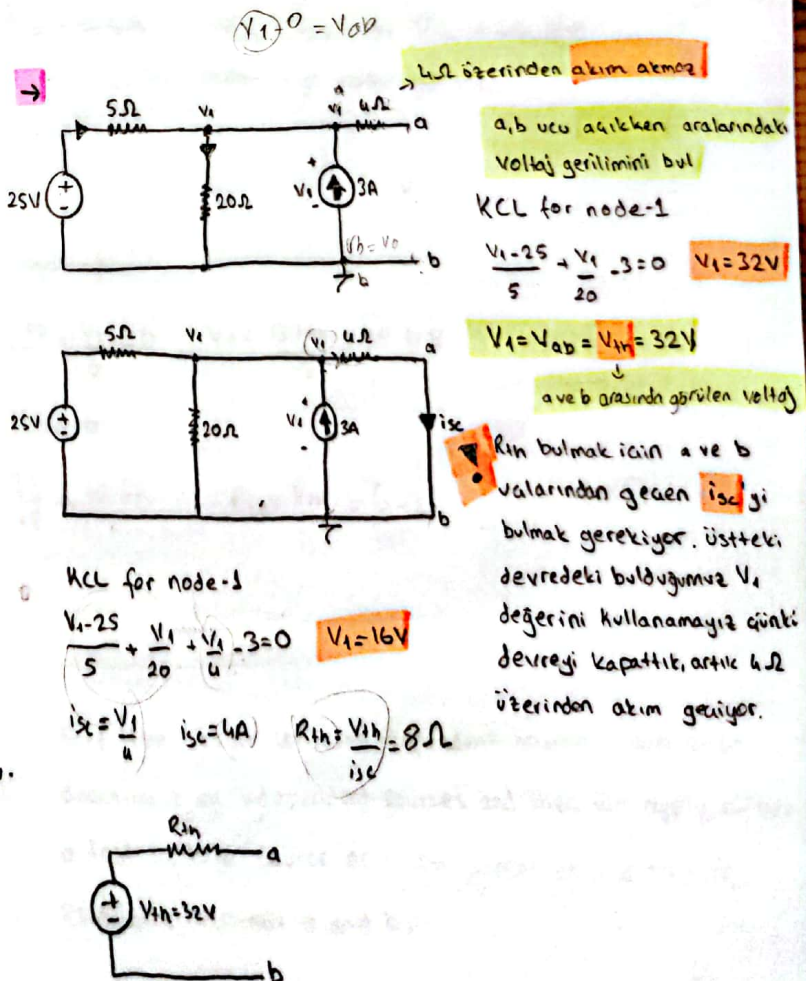
→ The Norton equivalent circuit consists of an independent current source in parallel with the Norton equivalent resistance R_N .

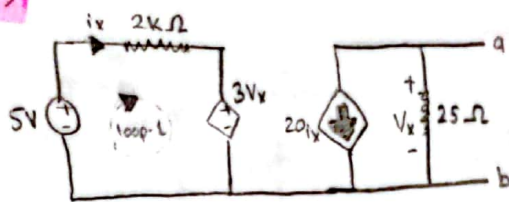


$$R_N = R_{th}, I_N = i_{sc}$$



→ Gücü sorulan kaynağı transformation ile yok etmemek lazım. Herhangi bir şey hesaplanırsa devrenin ilk halinden halletmekte fayda var.





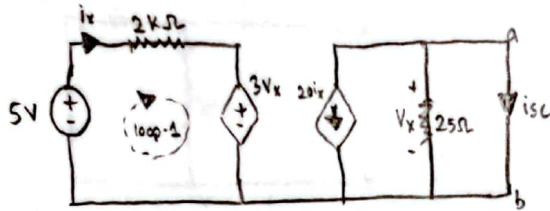
$$V_{th} = V_{ab} = V_x = -20 \cdot i_x \cdot 25 = -500 i_x \rightarrow \text{Seriler}$$

KVL for loop-1

$$-5 + 2000 i_x + 3V_x = 0 \quad i_x = \frac{1}{100} \text{ A}$$

$$V_{th} = V_{ab} = V_x = -5 \text{ V}$$

now we should find i_{sc} ?



$$V_{ab} = V_x = 0 \rightarrow V_a = V_b \rightarrow \text{Kısa devre} \rightarrow \frac{V_a - V_b}{25} = V_x, \quad V_x = 0$$

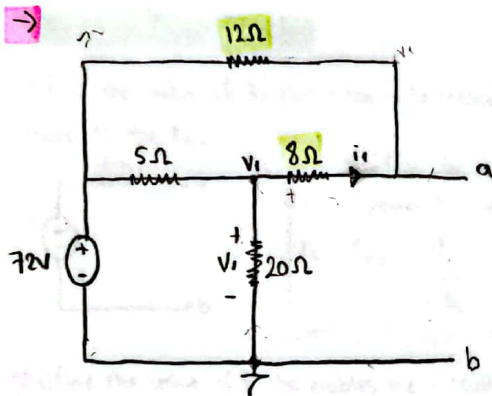
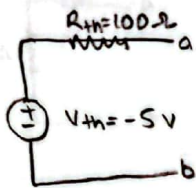
KVL for loop-1

$$-5 + 2000 i_x + 3V_x = 0 \quad i_x = \frac{5}{2000}$$

KCL for node a

$$20 i_x + i_{sc} = 0 \quad i_{sc} = -0.05 \text{ A}$$

$$R_{th} = \frac{V_{th}}{i_{sc}} = \frac{-5}{-0.05} = 100 \Omega$$



KCL for node-1

$$\frac{V_1}{20} + \frac{V_1 - 72}{5} + \frac{V_1 - 72}{8 + 12} = 0$$

Serilikten geliyor

$$V_1 = 60 \text{ V}$$

$$i_1 = \frac{V_1 - 72}{20} = -0.6 \text{ A}$$

$$V_{ab} = V_{th} = V_1 - 8i_1 = 64.8 \text{ V}$$

-den +'ya voltaj gidecek
↑
voltage düşecek

b'den a'ya kadar gidicez

$$+ i_{sc} + \frac{10 - V_1}{8} + \frac{0 - 72}{12}$$

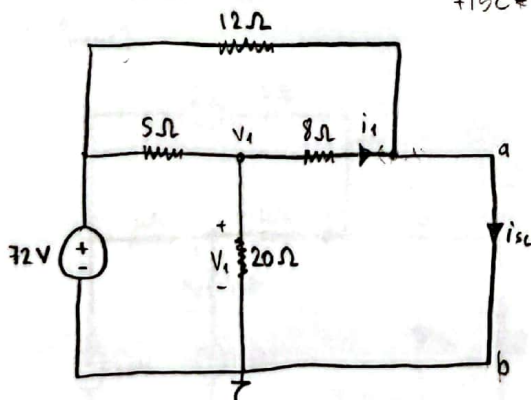
KCL for node-1

$$\frac{V_1}{20} + \frac{V_1 - 72}{5} + \frac{V_1}{8} = 0 \quad V_1 = \frac{576}{15} = 38.4 \text{ V}$$

KCL for node-a

$$i_{sc} = \frac{V_1}{8} + \frac{72}{12} = 10.8 \text{ A}$$

$$R_{th} = \frac{V_{th}}{i_{sc}} = 6 \Omega$$



More on Deriving a Thevenin Equivalent to find R_{th}

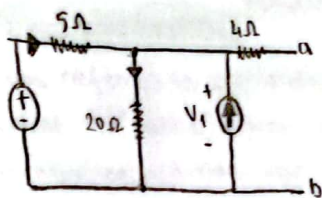
→ If the circuit contains **only** independent sources, we first deactivate all independent sources and calculate the equivalent resistance. (Replace each independent voltage source by a short circuit. Replace each independent current source by an open circuit.)

→ If the circuit contains **dependent** sources, we first deactivate all independent sources and then we apply either a test voltage source or a test current source to the Thevenin terminals a and b.

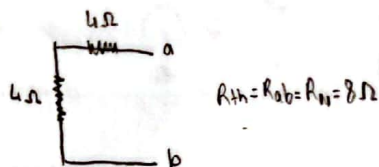
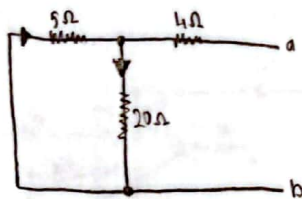
$$R_{th} = \frac{V_{ab}}{i_{ab}}$$

V_{ab} is the voltage rise from terminal b to terminal a

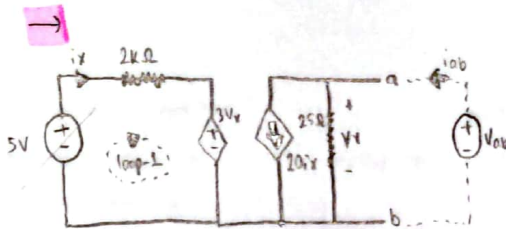
i_{ab} is the current flow from terminal b to terminal a



Deactivate all independent sources



$$R_{th} = R_{ab} = R_{th} = 8\Omega$$



$$R_{th} = \frac{V_{ab}}{i_{ab}}$$

KVL for loop-1

$$2000 \cdot i_x + 3V_x = 0$$

$V_{ab} = V_x$ because V_{ab} and 25Ω are connected in parallel

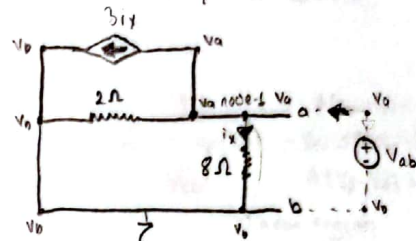
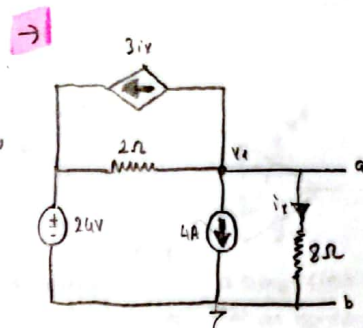
$$i_x = \frac{-3V_x}{2000} = \frac{-3V_{ab}}{2000}$$

KCL for node-a

$$i_{ab} = 20i_x + \frac{V_{ab}}{25}$$

$$i_{ab} = \frac{V_{ab}}{100}, R_{th} = 100\Omega$$

V_{ab} = 1 olarak alınabilir.



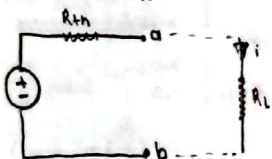
$$\text{Constraint equation: } i_x = \frac{V_{ab}}{8}$$

KCL for node-1

$$\frac{V_{ab}}{2} + \frac{V_{ab}}{8} + 3i_x - i_{ab} = 0 \quad R_{th} = 1\Omega$$

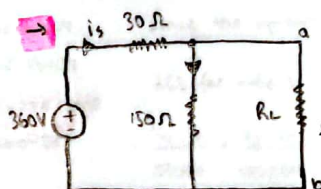
Maximum Power Transfer

→ Find the value of R_L that allow us to transfer maximum power to the R_L .

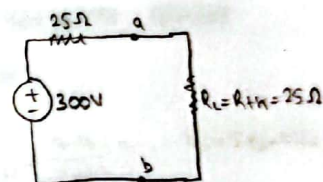


$R_L = R_{th}$ for maximum power transfer

$$P_{max} = \frac{V_{th}^2}{4R_L}$$



Source ⇒ transformation



$$P_{max} = P_{RL} = \frac{V_{th}^2}{4R_L} = 900W$$

a) Find the value of R that enables the circuit to deliver maximum power to the R

b) $P_{max} = ?$

a-1) What percentage of the power delivered by the 360V voltage source reaches R_L → R_L 'nin üzerinden GA'lik akım geçiriyor.

$$V_{ab} = 25 \cdot 6 = 150V$$

$$i_s = \frac{V_{ab}}{150} + \frac{V_{ab}}{25} = 7A$$

$P_{360} = 360 \cdot i_s = -2520W$ generates power
The percentage of the source power delivered to the load is: $\frac{900}{2520} \cdot 100 = 35.71\%$

$$\text{Constraint equation: } V_1 - 20 = V_0$$

KCL for node-1

$$\frac{V_1 - 100}{4} + \frac{V_1 - 20}{4} + \frac{V_1 - (V_0 + 100)}{8} = 0$$

$$V_1 = 80V$$

$$i_x = \frac{100 + V_0 - V_1}{8} = 10A$$

$$V_{th} = V_{ab} = V_1 + 4i_x = 80 + 4 \cdot 10 = 120V$$

$i_{sc} = ?$

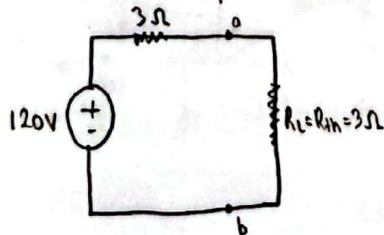
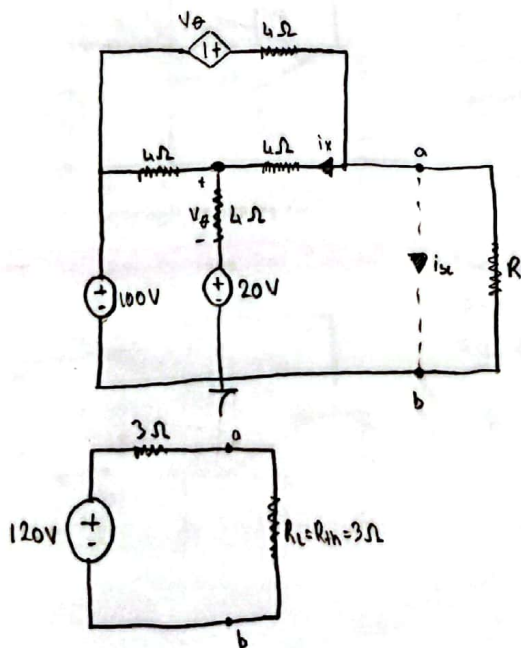
KCL for node-1

$$\frac{V_1 - 20}{4} + \frac{V_1}{4} + \frac{V_1 - 100}{4} = 0$$

$$V_1 = 40V \quad V_0 = 20V$$

KCL for node-a

$$i_{sc} = \frac{V_1}{4} + \frac{100 + V_0}{4} = 40A$$



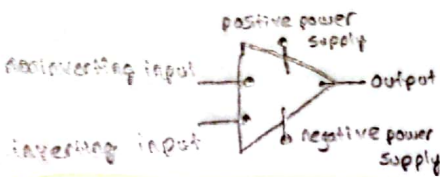
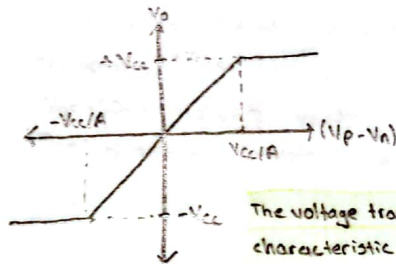
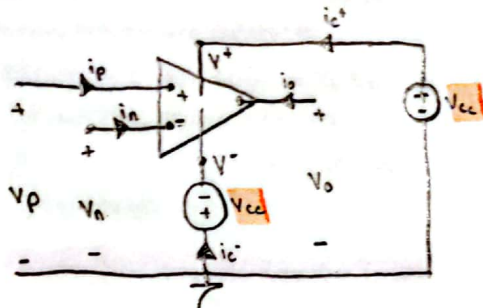
$$R_{th} = \frac{V_{th}}{i_{sc}} = \frac{120}{40} = 3\Omega = R \quad P_{max} = \frac{120^2}{12} = 1200W$$

CHAPTER 5

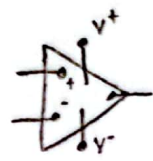
The Operational Amplifier

It was referred to **operational** because it was used to implement the mathematical operations of integration, differentiation, addition, sign changing, and scaling.

Gelen voltajı inputtan alıp, çıkışta outputtan çıkartır.



the circuit symbol for an operational amplifier (op-amp)



a simplified circuit symbol for an op-amp

$$V_o = \begin{cases} -V_{cc} & A(V_p - V_n) < -V_{cc} \\ A(V_p - V_n) & -V_{cc} \leq A(V_p - V_n) \leq V_{cc} \\ V_{cc} & A(V_p - V_n) > V_{cc} \end{cases}$$

linear region

Constraints:

$-A > 10^4$ for the most of op-amps

$|V_{cc}| \leq 20V$ for the most of op-amps

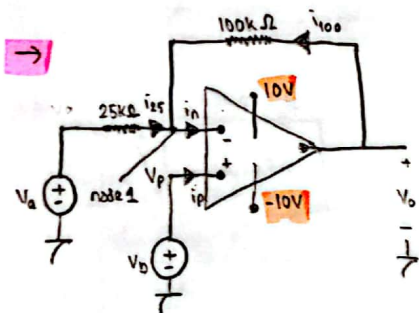
For the linear region:

$$|V_p - V_n| < \frac{20}{10^4} = 2mV$$

$$V_p \approx V_n$$

The equivalent resistance seen by the input terminals of the op-amp is very large, typically $1M\Omega$ or more.

$i_p = i_n \approx 0$ Using KCL: $i_p + i_{in} + i_c + i_c = 0 \Rightarrow i_o = -i_c - i_c$
For an ideal op-amp $A = \infty$, the input resistance $= \infty$



a) $V_o = ?$ If $V_a = 1V$ and $V_b = 0V$

b) $V_o = ?$ If $V_a = 1V$ and $V_b = 2V$

c) If $V_a = 1.5V$, specify the range of V_b that avoids amplifier saturation.

(The op-amp is ideal)

Since the op-amp is ideal $V_p = V_n = V_b$, $i_p = i_n = 0$

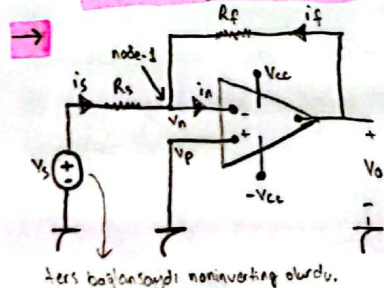
KCL for node-1 $i_{25} + i_{100} = i_n = 0$

$$\frac{V_a - V_n}{25000} + \frac{V_o - V_n}{100000} = 0 \Rightarrow \frac{V_a - V_b}{25000} + \frac{V_o - V_b}{100000} = 0 \Rightarrow V_o = 5V_b - 4V_a$$

a) $V_a = -4V$ b) $V_o = 6V$ c) $-10 \leq V_o \leq 10$ to work in the linear region $\Rightarrow -10 \leq 5V_b - 4V_a \leq 10$

If $-0.8V \leq V_b \leq 3.2V$, the op-amp will work in the linear region.

The Inverting-Amplifier Circuit



Açık devre kazançla noninverting devreye.

R_s : source resistor

R_f : feedback resistor

The op-amp is ideal;

$i_p = i_n = 0$, $V_p = V_n = 0$

$V_{cc} = 10V$ diğelim, $V_o \leftarrow$ 100 bölünebilir voltajdır.

KCL for node-1;

$$i_s + i_f = i_n = 0;$$

$$\frac{V_s - V_n}{R_s} + \frac{V_o - V_n}{R_f} = 0 \Rightarrow \frac{V_s}{R_s} + \frac{V_o}{R_f} = 0$$

$$V_o = -\frac{R_f}{R_s} V_s \quad \frac{R_f}{R_s} : \text{Gain}$$

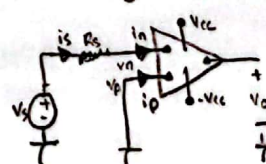
If we want the op-amp to work in the linear region

$$|V_o| \leq V_{cc} \Rightarrow \left| \frac{R_f}{R_s} V_s \right| \leq V_{cc} \Rightarrow \frac{R_f}{R_s} \leq \left| \frac{V_{cc}}{V_s} \right|$$

For example if $V_{cc} = 15V$ and maximum $V_s = 10mV$, $\frac{R_f}{R_s}$ must be less than 1500

to work in the linear region.

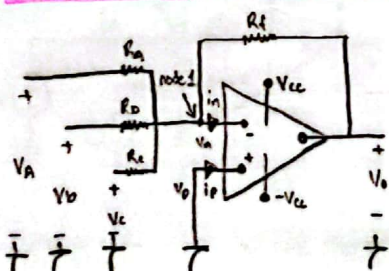
Opening the feedback.



$$V_o = -A \cdot V_n \approx -A \cdot V_o$$

An inverting amplifier operating open loop

The Summing-Amplifier Circuit



The op-amp is ideal

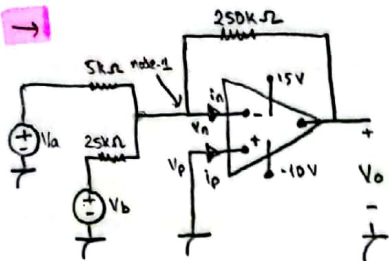
$$i_p = i_n = 0, \quad V_p = V_n = 0$$

KCL for node-1

$$\frac{V_a - V_n}{R_a} + \frac{V_b - V_n}{R_b} + \frac{V_c - V_n}{R_c} + \frac{V_o - V_n}{R_f} = i_n = 0$$

$$V_o = -\left(\frac{R_f}{R_a} V_a + \frac{R_f}{R_b} V_b + \frac{R_f}{R_c} V_c \right) \quad \text{If } R_a = R_b = R_c = R_s$$

Bütün akımları aynıdır
sıfırıya eşitler.



- a) $V_o = ?$ If $V_a = 0.1V$ and $V_b = 0.25V$
 b) If $V_b = 0.25V$, how large can V_a be before the op-amp saturates?
 c) If $V_a = 0.1$, how large can V_b be before the op-amp saturates?
 d) Repeat a, b, c with the polarity of V_b reversed.

$$a) V_o = -\left(\frac{250}{25}V_a + \frac{250}{25}V_b\right) = -50V_a - 10V_b$$

$$V_o = -7.5V$$

$$b) V_o = -50V_a - 10 \cdot 0.25 = -10 \quad -50V_a = -7.5$$

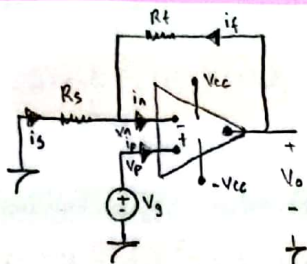
$$V_a = 0.15V \quad \text{The op-amp will saturate if } V_a \geq 0.15V$$

$$c) V_o = -50 \cdot 0.1 - 10V_b = -10 \quad -10V_b = -5$$

$$V_b = 0.5V \quad \text{The op-amp will saturate if } V_b \geq 0.5V$$

d) i) $V_o = -50V_a + 10V_b$, $V_o = -2.5V$
 ii) $V_o = -50V_a + 10 \cdot 0.25 = -10$, $V_a = 0.25V$
 The op-amp will saturate if $V_a \geq 0.25V$
 iii) $V_o = -50 \cdot 0.1 + 10V_b = 10$, $V_b = 2V$
 The op-amp will saturate if $V_b \geq 2V$

→ The Noninverting-Amplifier Circuit



The op-amp is ideal;

$$i_p = i_n = 0, V_n = V_p = V_i$$

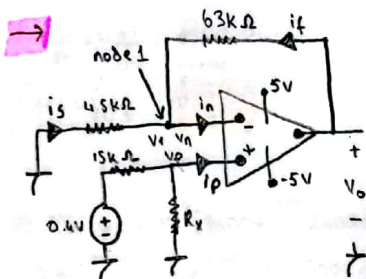
KCL for node-1

$$\frac{V_n}{R_s} + \frac{V_n - V_o}{R_f} = 0 \Rightarrow \frac{V_i}{R_s} + \frac{V_i - V_o}{R_f} = 0$$

$$V_o = V_i \frac{R_s + R_f}{R_s}$$

Operating in the linear region requires that

$$\frac{R_s + R_f}{R_s} < \left| \frac{V_{cc}}{V_i} \right|$$



The op-amp is ideal;

$$i_p = i_n = 0, V_n = V_p = V_i$$

Voltage divider; V_p, R_x varies under V_o constraint

$$V_p = 0.4 \times \frac{60k\Omega}{60k\Omega + 15k\Omega} = 0.32V = V_i$$

a) KCL for node-1

$$\frac{V_i}{4.5k\Omega} + \frac{V_i - V_o}{63k\Omega} = 0 \Rightarrow V_o = \frac{67.5}{4.5} \cdot V_i = 4.8V$$

$$b) V_p = 0.4 \cdot \frac{R_x}{R_x + 15k\Omega} = V_i$$

KCL for node-1

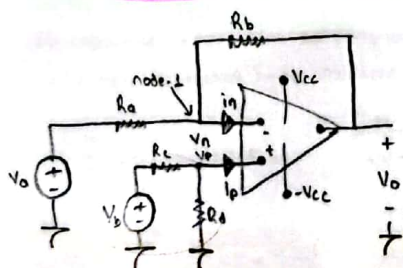
$$\frac{V_i}{4.5k\Omega} + \frac{V_i - V_o}{63k\Omega} = 0 \quad V_o = \frac{67.5}{4.5} \cdot V_i$$

$$V_o = \frac{67.5}{4.5} \cdot 0.4 \cdot \frac{R_x}{R_x + 15k\Omega}$$

The op-amp will saturate when V_o reaches to 5V.

$$V_o = 6 \cdot \frac{R_x}{R_x + 15k\Omega} = 5V \quad R_x = 75k\Omega$$

→ The Difference Amplifier Circuit



The op-amp is ideal; $i_p = i_n = 0, V_n = V_p = V_i$

Voltage divider

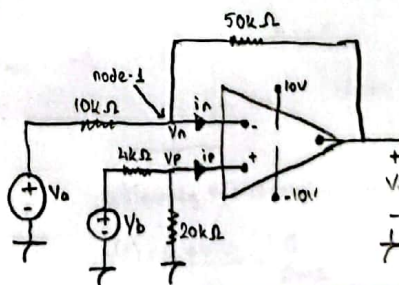
$$V_p = \frac{R_d}{R_d + R_c} \cdot V_b = V_i$$

KCL for node-1

$$\frac{V_i - V_a}{R_a} + \frac{V_i - V_o}{R_b} = 0 \quad V_o = V_i \left(\frac{R_b}{R_a} + 1 \right) - V_a \frac{R_b}{R_a}$$

$$V_o = \frac{R_d(R_a + R_b)}{R_a(R_c + R_d)} \cdot V_b - \frac{R_b}{R_a} V_a$$

$$\text{If } \frac{R_a}{R_b} = \frac{R_c}{R_d}, \quad V_o = \frac{R_b}{R_a} (V_b - V_a)$$



The op-amp is ideal; $i_p = i_n = 0, V_n = V_p = V_i$

$$V_p = \frac{20}{20+40} V_b, \quad \frac{5V_b}{6} = V_i$$

a) KCL for node-1

$$\frac{V_i - V_a}{10k\Omega} + \frac{V_i - V_o}{50k\Omega} = 0, \quad V_o = 6V_i - 5V_a$$

$$V_o = 5(V_b - V_a), \quad \text{if } V_b = 4V, \quad V_o = 5(4 - V_a)$$

$$-10 \leq V_o \leq 10, \quad 2 \leq V_a \leq 6$$

The op-amp will operate in the linear region if $2 \leq V_a \leq 6$.

a) What range values for V_a will result in linear operation if $V_b = 4V$?

b) Repeat (a) with 20kΩ resistor decreased to 8kΩ

$$b) V_p = \frac{8}{8+40} V_b = V_i$$

$$V_o = 6V_i - 5V_a = 4V_b - 5V_a$$

$$\text{If } V_b = 4V, \quad V_o = 16 - 5V_a$$

$$-10 \leq V_o \leq 10, \quad -10 \leq 16 - 5V_a \leq 10$$

$$1.2 \leq V_a \leq 5.2$$

The op-amp will operate in the linear region if $1.2 \leq V_a \leq 5.2$

CHAPTER-6

Inductance, Capacitance and Mutual Inductance

The Inductor

Göbin

→ An inductor is a passive element designed to store energy in its magnetic field. They are used in power supplies, transformers, radios, TV's, radars, and electric motors.

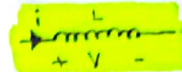
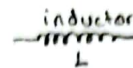
→ An inductor is composed of a coil of wire wound around a supporting core whose material may be magnetic or nonmagnetic.

→ If the current is varying with time, the magnetic field is varying with time. A time varying magnetic field induces a voltage in any conductor linked by the field.

→ Current in an inductor in term of the voltage across the inductor

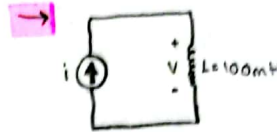
$$V = L \frac{di}{dt} \Rightarrow V dt = L \frac{di}{dt} dt \Rightarrow V dt = L di \Rightarrow L \int_{i(t_0)}^{i(t)} dv = \int_{t_0}^t V(\tau) d\tau$$

$$\Rightarrow i(t) = \frac{1}{L} \int_{t_0}^t V(\tau) d\tau + i(t_0)$$



Inductance in Henry

$$V = L \frac{di}{dt}$$



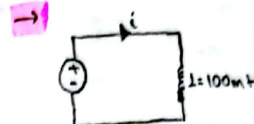
$$i = 0 \text{ A for } t \leq 0$$

$$i = 10t e^{-5t} \text{ A for } t \geq 0$$

$$V(t) = ?$$

$$V(t) = L \frac{di(t)}{dt}$$

$$= 0.1 \times \frac{d(10t e^{-5t})}{dt} = e^{-5t} - 5t e^{-5t} = e^{-5t} (1 - 5t) \text{ V}$$



$$V = 0 \text{ V for } t \leq 0$$

$$V = 20t e^{-10t} \text{ V for } t \geq 0$$

$$i = 0 \text{ for } t \leq 0$$

$$i(t) = ?$$

$$i(t) = \frac{1}{L} \int_{t_0}^t V(\tau) d\tau + i(t_0) = \frac{1}{0.1} \int_0^t 20\tau e^{-10\tau} d\tau + 0$$

$$i(t) = 200 \left[-\frac{e^{-10\tau}}{100} (10\tau + 1) \right]_0^t$$

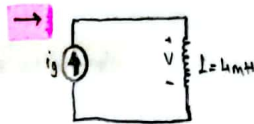
$$= 2(1 - 10t e^{-10t} - e^{-10t}) \text{ A for } t \geq 0$$

Power and energy in the inductor

$$P = V \cdot i = L i \frac{di}{dt} = V \cdot \left[\frac{1}{L} \int_{t_0}^t V(\tau) d\tau + i(t_0) \right]$$

$$P = \frac{dW}{dt} = L i \frac{di}{dt} \quad dW = L i di$$

$$\int_0^W dW = L \int_0^i i di \quad W = \frac{1}{2} L i^2$$



$$i(t) = 0 \text{ A for } t \leq 0$$

$$i(t) = 8e^{-300t} - 8e^{-1200t} \text{ A for } t \geq 0$$

$$a) V(t) = L \frac{di}{dt} = 4 \times 10^{-3} [9600e^{-300t} - 2400e^{-1200t}]$$

$$V(0) = 4(9.6 - 2.4) = 28.8 \text{ V}$$

$$b) V(t) = 0 = 4 \cdot 10^{-3} [9600e^{-300t} - 2400e^{-1200t}]$$

$$9.6e^{-1200t} = 2.4e^{-300t} \Rightarrow -300t = \ln \frac{2.4}{9.6} \quad t = 1.54 \text{ msec}$$

$$d) \frac{dP(t)}{dt} = 0 = e^{-600t} [46080 - 576000e^{-300t} + 737280e^{-1200t}] = 0$$

sıfır oldu

$$46080 - 576000y + 737280y^2 = 0 \quad t = 1.11.05 \text{ msec}$$

$$e) P_{\max} = P(1.11.05 \text{ msec}) = 32.72 \text{ W}$$

$$a) V(0) = ?$$

b) The instant of time, greater than zero, when the voltage V passes through zero?

$$c) P(t) = ?$$

d) The instant when the power delivered to the inductor is maximum?

e) The maximum power for the inductor?

f) The maximum energy stored in the inductor?

g) The instant of time when the stored energy is maximum?

$$c) P(t) = V(t) \cdot i(t) = 4[9.6e^{-1200t} - 2.4e^{-300t}] 8[e^{-300t} - e^{-1200t}] \text{ A for } t \geq 0$$

$$P(t) = 76.8e^{-600t} + 384e^{-1500t} - 307.2e^{-2400t}$$

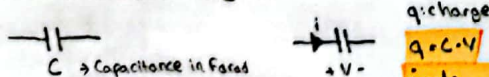
g) The instant of time = 1.54 msec (the energy will be max when the current is max)

Akım ne zaman max olur? Akım formülünün türevini alıp sıfıra alırsak ulaşırız. B şıkkinda birer öyle yapmış oldu $V=0$ den i max oldu

$$f) W = \frac{1}{2} L i^2 = \frac{1}{2} 4 \cdot 10^{-3} \cdot i^2(1.54 \text{ msec}) \quad W_{\max} = 28.57 \text{ mJ}$$

The Capacitor

→ A capacitor is an electrical component that consists of two conductors separated by an insulator or dielectric material.



q: charge

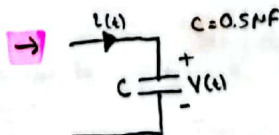
$$q = C \cdot V$$

$$i = \frac{dq}{dt} = C \cdot \frac{dV}{dt}$$

$$V(t) = \frac{1}{C} \int_{t_0}^t i(\tau) d\tau + V(t_0)$$

$$W = \frac{1}{2} C V^2$$

$$P = V \cdot i = C V \frac{dV}{dt} = i \left[\frac{1}{C} \int_{t_0}^t i(\tau) d\tau + V(t_0) \right]$$



$$V(t) = \begin{cases} 0 & t < 0 \\ 4t & 0 \leq t \leq 1 \text{ sec} \\ 4e^{-(t-1)} & t \geq 1 \text{ sec} \end{cases}$$

a) $i(t) = ?$, $P(t) = ?$, $W(t) = ?$
b) Specify the interval of time when energy is being stored in the capacitor.

c) Specify the interval of time when energy is being delivered by the capacitor.

$$i(t) = C \frac{dV(t)}{dt} = \begin{cases} 0 & t < 0 \\ 2 \text{ mA} & 0 \leq t \leq 1 \text{ sec} \\ -2e^{-(t-1)} \text{ mA} & t \geq 1 \text{ sec} \end{cases}$$

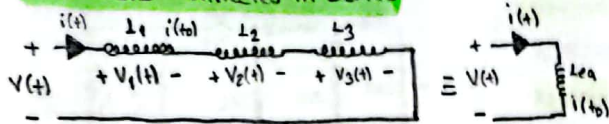
$$P(t) = i(t) \cdot V(t) = \begin{cases} 0 & t < 0 \\ 8t \text{ mW} & 0 \leq t \leq 1 \text{ sec} \\ 9e^{-2(t-1)} \text{ mW} & t \geq 1 \text{ sec} \end{cases}$$

$$W(t) = \frac{1}{2} C V^2(t) = \begin{cases} 0 & t < 0 \\ 4t^2 \text{ mJ} & 0 \leq t \leq 1 \text{ sec} \\ 4e^{-2(t-1)} \text{ mJ} & t \geq 1 \text{ sec} \end{cases}$$

b) The energy is stored if the power is positive. The energy is being stored in the interval 0-1 sec

c) The energy is being delivered by the capacitor whenever the power is negative. The energy is being delivered for all t greater than 1 sec

Inductors Connected in Series

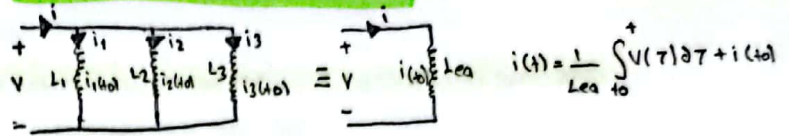


$$V_1(t) = L_1 \frac{di(t)}{dt}, V_2(t) = L_2 \frac{di(t)}{dt}, V_3(t) = L_3 \frac{di(t)}{dt} \quad V(t) = L_{eq} \frac{di(t)}{dt}$$

$$V(t) = V_1(t) + V_2(t) + V_3(t) = (L_1 + L_2 + L_3) \frac{di(t)}{dt} = L_{eq} \frac{di(t)}{dt}$$

$$L_{eq} = L_1 + L_2 + L_3$$

Inductors Connected in Parallel



$$i_1(t) = \frac{1}{L_1} \int_0^t V(\tau) d\tau + i_1(t_0)$$

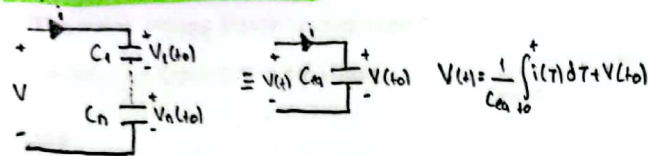
$$i_2(t) = \frac{1}{L_2} \int_0^t V(\tau) d\tau + i_2(t_0)$$

$$i_3(t) = \frac{1}{L_3} \int_0^t V(\tau) d\tau + i_3(t_0)$$

$$i(t) = i_1(t) + i_2(t) + i_3(t) = \left(\frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} \right) \int_0^t V(\tau) d\tau + i_1(t_0) + i_2(t_0) + i_3(t_0)$$

$$\frac{1}{L_{eq}} = \left(\frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} \right), \quad i(t_0) = i_1(t_0) + i_2(t_0) + i_3(t_0)$$

Capacitors Connected in Series

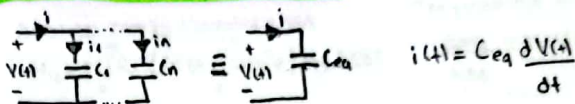


$$V(t) = V_1(t) + V_2(t) = \frac{1}{C_1} \int_0^t i(\tau) d\tau + V_1(t_0) + \frac{1}{C_2} \int_0^t i(\tau) d\tau + V_2(t_0)$$

$$= \left(\frac{1}{C_1} + \frac{1}{C_2} \right) \int_0^t i(\tau) d\tau + V_1(t_0) + V_2(t_0)$$

$$\frac{1}{C_{eq}} = \left(\frac{1}{C_1} + \frac{1}{C_2} \right), \quad V(t_0) = V_1(t_0) + V_2(t_0)$$

Capacitors Connected in Parallel

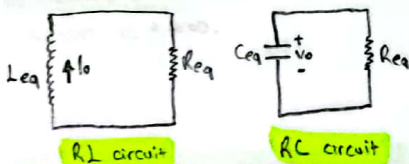


$$i(t) = i_1(t) + i_2(t) = C_1 \frac{dV(t)}{dt} + C_2 \frac{dV(t)}{dt} = (C_1 + C_2) \frac{dV(t)}{dt}$$

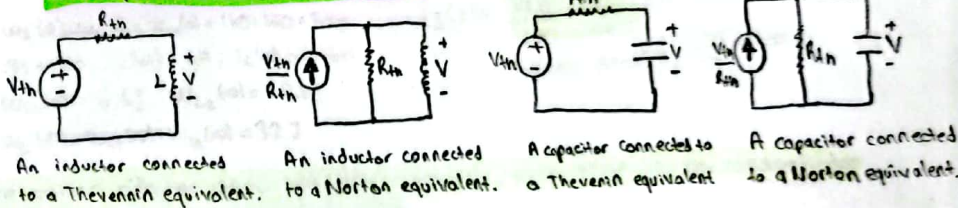
$$C_{eq} = C_1 + C_2$$

CHAPTER 7

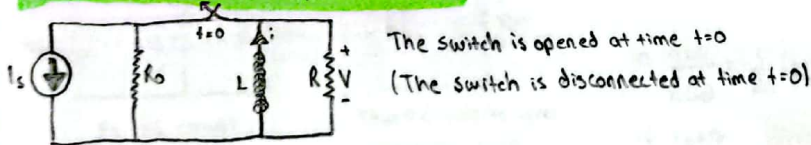
Response of First Order RL and RC Circuits



Four possible first-order circuits

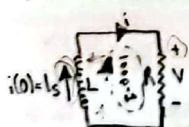


The Natural Response of an RL Circuit



$$i(0^-) = I_s$$

$$i(0^+) = i(0^-) = I_s \quad \text{Inductor current can not be changed immediately}$$



KVL for loop-1

$$L \frac{di}{dt} + Ri = 0 \Rightarrow \frac{di}{i} = -\frac{R}{L} dt \Rightarrow \int_{i(t_0)}^i \frac{dx}{x} = -\frac{R}{L} \int_{t_0}^t dy$$

$$\ln \frac{i(t)}{i(t_0)} = -\frac{R}{L} (t - t_0) \quad i(t) = i(t_0) e^{-\frac{R}{L} (t - t_0)} \quad \text{for } t \geq t_0$$

$$V = Ri = I_s R e^{-\frac{R}{L} t} \quad \text{for } t \geq 0^+ \quad V(0^-) = 0 \quad V = L \frac{di}{dt} = 0 \quad \text{for } t \leq 0$$

The power dissipated in the resistor

$$P = Vi = i^2 R = I_s^2 R e^{-\frac{2R}{L} t} \quad t \geq 0^+$$

The energy delivered to the resistor during interval of time after the switch has been opened is

$$W = \int_0^t P(x) dx = \int_0^t I_s^2 R e^{-\frac{2R}{L} x} dx = \frac{L R I_s^2}{2R} (1 - e^{-\frac{2R}{L} t}) = \frac{1}{2} L I_s^2 (1 - e^{-\frac{2R}{L} t}) \quad t \geq 0^+$$

The significance of the time constant

$$\tau = \text{time constant} = \frac{L}{R} \quad \text{time constant for the RL circuit}$$

$$i(t) = I_s e^{-\frac{t}{\tau}} = I_s e^{-\frac{R}{L} t} \quad \text{for } t \geq 0^+$$

$$V(t) = I_s R e^{-\frac{t}{\tau}} \quad \text{for } t \geq 0^+$$

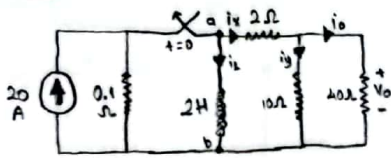
$$P = I_s^2 R e^{-\frac{2t}{\tau}} \quad \text{for } t \geq 0^+$$

$$W = \frac{1}{2} L I_s^2 (1 - e^{-\frac{2t}{\tau}}) \quad \text{for } t \geq 0^+$$

$$\text{If } t = 5\tau \Rightarrow i(t) \leq \frac{1}{100} i(0) \text{ and } V(t) \leq \frac{1}{100} V(0^+)$$

Thus, we sometimes say that, five time constants after switching, the current and voltages have, for most practical purposes, reached their final values.

useur bir size belavildikler
sonda
inductor kisa devre
capacitor uzun devre
gibi davranir.



The switch has been closed for a long time before it is opened at time $t=0$.

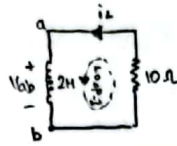
- a) $i_L(t)$ for $t \geq 0$ b) $i_o(t)$ for $t \geq 0^+$
c) $V_o(t)$ for $t \geq 0^+$ d) The percentage of the total energy stored in 2H inductor that is dissipated in the 10Ω resistor.

inductor kısa devre olarak davranacağından bütün 20A akımı onun üzerinden geçer

$$i_L(0^-) = i_L(0^+) = 20A$$

Anahtar açılacağından devrenin sol tarafı ile ilgilenmemize gerek yok

$$R_{eq} = 2 + \frac{40 \cdot 10}{40+10} = 10\Omega \text{ for } t \geq 0^+$$



KVL for loop-1

$$L \frac{di_L}{dt} + R i_L = 0 \Rightarrow i_L(t) = i_L(0) e^{-\frac{R}{L}t} = 20 e^{-5t} \text{ for } t \geq 0$$

$$V_o = L \frac{di_L}{dt} = -200 e^{-5t} \text{ for } t \geq 0$$

$$i_y = \frac{V_o}{10} = -20 e^{-5t} \text{ for } t \geq 0^+$$

Current divider circuit

$$i_o(t) = \frac{10}{10+40} i_y = -4 e^{-5t} \text{ for } t \geq 0^+$$

$$V_o(t) = 40 i_o(t) = -160 e^{-5t} \text{ for } t \geq 0^+$$

The total energy stored in L at time 0:

$$W_L(0) = \frac{1}{2} L i_L^2(0) = 0.5 \cdot 2 \cdot 20^2 = 400J$$

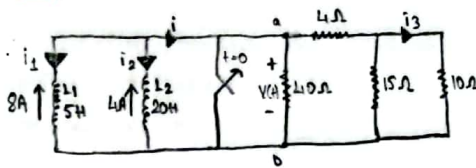
$$i_y = \frac{40}{10+40} i_L = -16 e^{-5t} \text{ for } t \geq 0^+$$

The total energy dissipated in 10Ω

$$W_{10\Omega} = \int_0^\infty R i_y^2 dt = \int_0^\infty 10 \cdot 16^2 e^{-10t} dt = 256J$$

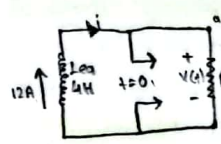
The percentage of energy dissipated in the 10Ω

$$\frac{256}{400} \cdot 100 = 64\%$$



The initial currents in inductors L_1 and L_2 have been established by sources that is not shown.

- a) $i_1(t)$, $i_2(t)$, $i_3(t)$, for $t \geq 0$
b) Calculate the initial energies stored in inductors.
c) Determine how much energy stored in the inductors as $t \rightarrow \infty$.



$$R_{eq} = 20\Omega \parallel (4\Omega + (15\Omega \parallel 10\Omega)) = 8\Omega$$

$$L_{eq} = \frac{20 \cdot 5}{20+5} = 4H$$

$$V(t) = R_{eq} \cdot i(t) = 96 e^{-2t} \text{ for } t \geq 0$$

$$i_1(t) = \frac{1}{L_1} \int_0^t V(\tau) d\tau + i_1(0) = 1.6 - 9.6 e^{-2t} \text{ for } t \geq 0$$

$$i_2(t) = \frac{1}{L_2} \int_0^t V(\tau) d\tau + i_2(0) = -1.6 - 2.4 e^{-2t} \text{ for } t \geq 0$$

$$i_3(t) = \frac{V(t) \cdot 15}{10+15} = 5.76 e^{-2t} \text{ for } t \geq 0$$

current divider, $\frac{V(t)}{10}$ akımı.

$$W_{L1}(0) = \frac{1}{2} L_1 i_1^2(0) = 0.5 \cdot 5 \cdot 64 = 160J$$

$$W_{L2}(0) = \frac{1}{2} L_2 i_2^2(0) = 0.5 \cdot 20 \cdot 16 = 160J$$

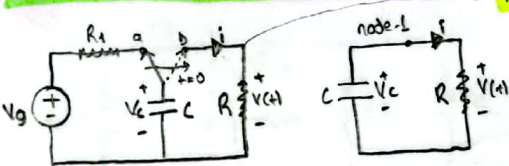
$$W_L(0) = W_{L1}(0) + W_{L2}(0) = 160 + 160 = 320J$$

$$As \rightarrow \infty \quad i_1(\infty) = 1.6A, \quad i_2(\infty) = -1.6A$$

$$W_{L1}(\infty) = 6.4J \quad W_{L2}(\infty) = 25.6J$$

$$W_L(\infty) = W_{L1}(\infty) + W_{L2}(\infty) = 32J$$

Natural Response of an RC Circuit



An RC circuit

The RC Circuit after switching ($t > 0$)

Özellikle, anahtar açıldığında akım kaldığında, Capacitor akım devre gibi davranacağından R_1 üzerinden akım geçmeyecek, $V_o = V_C$ olacak

$$KCL \text{ for node-1} \quad C \frac{dV_C}{dt} + \frac{V_C}{R} = 0$$

$$\frac{dV_C}{V_C} = -\frac{1}{RC} dt \quad \int_{V_C(0)}^{V_C(t)} \frac{dV_C}{V_C} = -\frac{1}{RC} \int_0^t dt$$

$$\ln \frac{V_C(t)}{V_C(0)} = -\frac{1}{RC} (t-t_0) \quad V_C(t) = V_C(t_0) e^{-\frac{1}{RC} (t-t_0)}$$

$$\text{If } t_0 = 0 \quad V_C(t) = V_C(0) e^{-\frac{t}{RC}} \text{ for } t \geq 0$$

$$V_C(0^-) = V_C(0^+) = V_C(0) = V_0 = V_o$$

$$\tau = RC$$

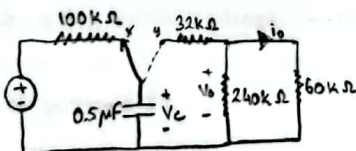
$$V_C(t) = V_o e^{-\frac{t}{\tau}} \text{ for } t \geq 0$$

$$i(t) = \frac{V_C(t)}{R} = \frac{V_o}{R} e^{-\frac{t}{\tau}} \text{ for } t \geq 0$$

$$P = V_C(t) \cdot i(t) = \frac{V_o^2}{R} e^{-\frac{2t}{\tau}} \text{ for } t \geq 0$$

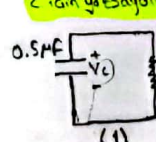
devre için güç kaybı
C için güç kaybolmuş - discharge

$$W = \int_0^\infty P dt = \int_0^\infty \frac{V_o^2}{R} e^{-\frac{2t}{\tau}} dt = \frac{1}{2} C V_o^2 (1 - e^{-\frac{2t}{\tau}}) \text{ for } t \geq 0$$



The switch has been in position x for a long time. At time $t=0$, the switch moves instantaneously to position y.

- a) $V_C(t)$? for $t \geq 0$.
b) $V_o(t)$? for $t \geq 0$.
c) $i_o(t)$? for $t \geq 0$.
d) The total energy dissipated in the 60k resistor?



$$(1) \quad V_C(0^+) = V_C(0^-) = V_C(0) = 100V$$

$$\tau = C \cdot R_{eq} = 0.5 \cdot 10^{-6} \cdot 80 \cdot 10^3 = 40 \mu\text{sec} = 40 \cdot 10^{-3} \text{ sec}$$

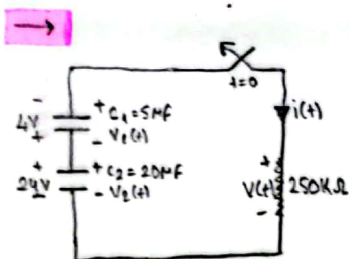
$$(1) \quad V_C(t) = V_C(0) e^{-\frac{t}{\tau}} = 100 e^{-\frac{t}{40 \cdot 10^{-3}}}$$

$$(3) \quad V_o(t) = \frac{48}{48+32} \cdot V_C(t) = 60 e^{-\frac{t}{40 \cdot 10^{-3}}} \text{ (Voltage divider circuit)}$$

$$(2) \quad i_o(t) = \frac{V_o(t)}{60k\Omega} = e^{-\frac{t}{40 \cdot 10^{-3}}} \text{ mA}$$

$$W = \int_0^\infty i_o^2 R dt = \int_0^\infty 60 \cdot 10^3 \cdot e^{-\frac{2t}{40 \cdot 10^{-3}}} dt = 1.2 \cdot 10^3 \text{ joules}$$

0.1A



The initial voltages on capacitors C_1 and C_2 have been established by sources that is not shown here.

The switch is closed at $t=0$.

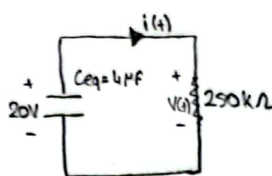
a) $V_1(t)$, $V_2(t)$ for $t \geq 0$ and

$i(t)$ for $t \geq 0$

b) Calculate the initial energies stored in C_1 and C_2 .

c) Determine how much energy is stored in capacitors as $t \rightarrow \infty$.

d) Show that the total energy delivered to the $250k\Omega$ resistor is the difference between the result obtained in (b) and (c).



$$V_1(0^+) = V_1(0^-) = 4V \quad V_2(0^+) = V_2(0^-) = 20V \quad V(0^+) = V_1(0^+) + V_2(0^+) = 20V$$

$$C_{eq} = \frac{5 \cdot 20}{5 + 20} = 4\mu F \quad T = C_{eq} \cdot R = 4 \cdot 10^{-6} \cdot 250 \cdot 10^3 = 1 \text{ sec}$$

$$V(t) = V(0) e^{-\frac{t}{T}} = 20 e^{-t} \text{ for } t \geq 0$$

$$i(t) = \frac{V(t)}{250k} = 80 e^{-t} \mu A$$

$$V_1(t) = \frac{1}{C_1} \int_0^t i(\tau) d\tau + V_1(0) = \frac{10^{-6}}{5} \int_0^t 80 \cdot 10^6 e^{-\tau} d\tau - 4 = 16 e^{-t} - 20V$$

$$V_2(t) = \frac{1}{C_2} \int_0^t i(\tau) d\tau + V_2(0) = \frac{10^{-6}}{20} \int_0^t 80 \cdot 10^6 e^{-\tau} d\tau + 20 = 4 e^{-t} + 20V$$

b) $W_{C1}(0) = 0.5 \cdot C_1 \cdot V_1^2(0) = 0.5 \cdot 5 \cdot 10^{-6} \cdot 4^2 = 40 \mu J$ The total energy stored in the two capacitors is: $W(0) = W_{C1}(0) + W_{C2}(0) = 5800 \mu J$
 $W_{C2}(0) = 0.5 \cdot C_2 \cdot V_2^2(0) = 0.5 \cdot 20 \cdot 10^{-6} \cdot 20^2 = 5760 \mu J$

c) As $t \rightarrow \infty$ $V_1 \rightarrow -20$, $V_2 \rightarrow 20$

$$W_{C1}(\infty) = 0.5 \cdot C_1 \cdot V_1^2(\infty) = 1000 \mu J \quad W_{C2}(\infty) = 0.5 \cdot C_2 \cdot V_2^2(\infty) = 4000 \mu J$$

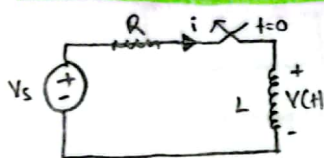
$$W(\infty) = W_{C1}(\infty) + W_{C2}(\infty) = 5000 \mu J$$

$$d) W_{250k\Omega} = \int_0^{\infty} i^2 R dt = \int_0^{\infty} 6400 e^{-2t} \cdot 10^{-12} \cdot 250 \cdot 10^3 dt = 800 \mu J$$

$$\text{From part b and c} \quad W(0) = 5800 \mu J, \quad W(\infty) = 5000 \mu J$$

$$W(0) - W(\infty) = 800 \mu J = W_{250k\Omega}$$

The Step Response of RL Circuits



$$i(t) = \frac{V_s}{R} + \left(i(t_0) - \frac{V_s}{R} \right) e^{-\frac{R}{L}(t-t_0)} \quad t \geq t_0$$

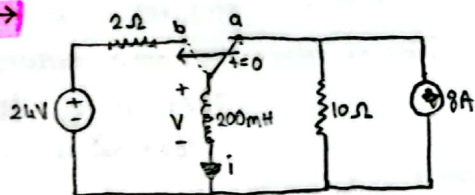
if $i(t_0) = 0$ and $t_0 = 0$

$$i(t) = \frac{V_s}{R} - \frac{V_s}{R} e^{-\frac{R}{L}t} \quad t \geq 0$$

$$V(t) = (V_s - R i(t_0)) e^{-\frac{R}{L}(t-t_0)} \quad t \geq t_0$$

if $i(t_0) = 0$ and $t_0 = 0$

$$V(t) = V_s e^{-\frac{R}{L}t} \quad t \geq 0$$



The switch has been position "a" for a long time. At $t=0$, the switch moves from position "a" to position "b".

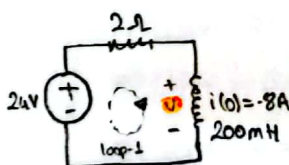
a) Find $i(t)$ for $t \geq 0$.

b) What is the initial voltage across the inductor just after the switch has been moved to position "b".

c) How many milliseconds after the switch has moved to position "b" does the inductor voltage equal 24V?

$$i(0) = -8A$$

uzun süre bekletildiği için tüm 8A inductor üzerinden geçer.



$$a) i(t) = \frac{V_s}{R} + \left(i(0) - \frac{V_s}{R} \right) e^{-\frac{R}{L}t} = \frac{24}{2} + \left(-8 - \frac{24}{2} \right) e^{-\frac{2}{0.2}t}$$

$$i(t) = 12 - 20 e^{-10t} A \quad t \geq 0$$

b) KVL for loop-1

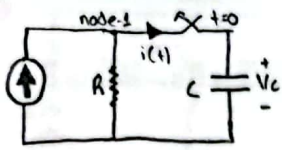
$$24 + 2i(0) = 0 \quad V = 24 - 2i(0) = 24 - 2(-8) = 40V$$

c) $V = 24 - 2i(t)$

$$24 = 24 - 2(12 - 20 e^{-10t})$$

$$t = 51.08 \text{ ms}$$

The Step Response of an RC Circuit



$$V_c(t) = RI_s + (V_c(t_0) - RI_s)e^{-\frac{1}{RC}(t-t_0)} \quad t \geq t_0$$

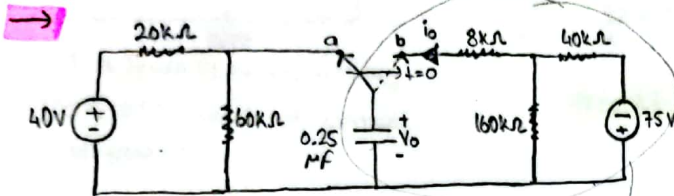
if $t_0 = 0$

$$V_c(t) = RI_s + (V_c(0) - RI_s)e^{-\frac{1}{RC}t} \quad t \geq 0$$

$$i(t) = C \frac{dV_c(t)}{dt}$$

$$i(t) = I_s - \frac{V_c(t)}{R} = \left(I_s - \frac{V_c(0)}{R} \right) e^{-\frac{1}{RC}t} \quad \text{for } t \geq 0^+$$

The switch has been closed at $t=0$.



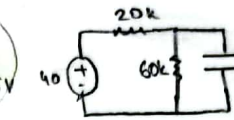
The switch has been in position "a" for a long time.

At $t=0$, the switch moves to position "b". Find

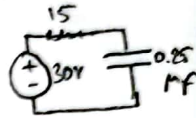
a) $V_o(t)$ for $t \geq 0$

b) $i_o(t)$ for $t \geq 0^+$

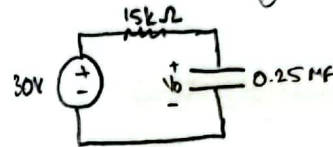
Source transformation



Source transformation

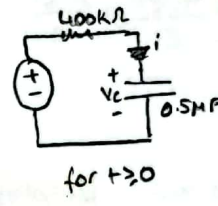
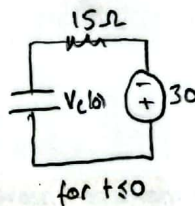
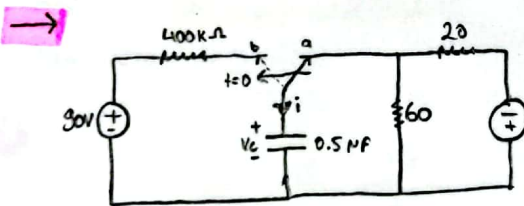
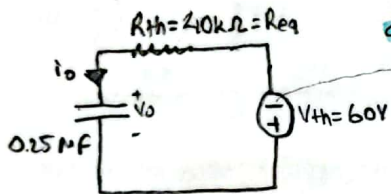


$$V_o(0) = 30V$$



a) $V_o(t) = RI_o + (V_o(t_0) - RI_o)e^{-\frac{1}{RC}t}$ for $t \geq 0$
 $= -60 + 90e^{-100t}$ for $t \geq 0$

b) $i_o(t) = C \frac{dV_o(t)}{dt} = -2.25e^{-100t}$ mA for $t \geq 0$



The switch has been in position "a" for a long time. At $t=0$, the switch moves to position "b".

a) what is the initial value of V_c ?

b) what is the final value of V_c ?

c) What is the time constant for $t \geq 0^+$?

d) $V_c(t)$ for $t \geq 0$?

e) $i(t)$ for $t \geq 0^+$?

f) How long after the switch is in position

"b" does the capacitor voltage equal zero?

a) $V_c(0) = -40 \cdot \frac{60}{60+20} = -30V$ voltage divider circuit

b) $V_c(\infty) = V_{cf} = 90V$

c) $\tau = RC = 0.2 \text{ sec}$

d) $V_c(t) = V_{cf} + [V_c(0) - V_{cf}]e^{-\frac{t-t_0}{\tau}} = 90 + (-30 - 90)e^{-\frac{t}{0.2}}$

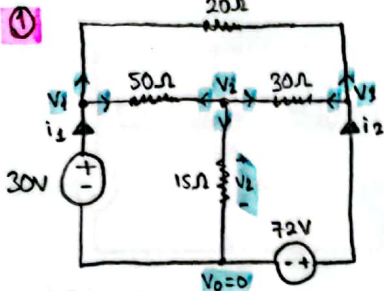
$$V_c(t) = 90 - 120e^{-5t} \quad t \geq 0$$

e) $i(t) = C \frac{dV_c}{dt} = \frac{V_{400k}}{400k} = 300e^{-5t}$ for $t \geq 0^+$

f) $V_c(t) = 0 = 90 - 120e^{-5t} \quad t = 0.057536 \text{ sec.}$

isa can Gabuk

CIRCUIT 2021 VİZE (AYNİ ZAMANDA 2022 VİZE İLK UĞ SORUSU)



Find i_1 and i_2 for the following circuit. (Use which ever method you want.)

Node-Voltage Method

$$V_1 = 30 \quad V_0 \rightarrow \text{reference node}$$

$$V_3 = 72$$

KCL for node-2

$$\frac{V_2 - 30}{50} + \frac{V_2}{15} + \frac{V_2 - 72}{30} = 0$$

$$V_2 = 25$$

(Devreye yasadıkları mı, nasıl ile ağız dım.)

KCL for node-1

$$-i_1 + \frac{V_1 - V_3}{20} + \frac{V_1 - V_2}{50} = 0$$

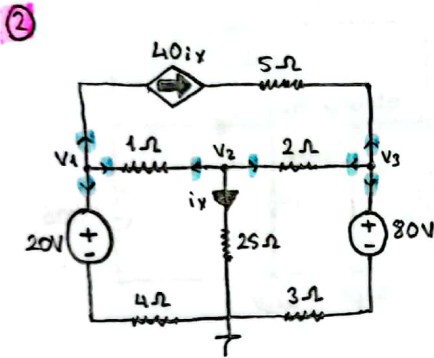
$$i_1 = -2$$

- olması yönünün ters olduğu anlamına geliyor sadece

KCL for node-3

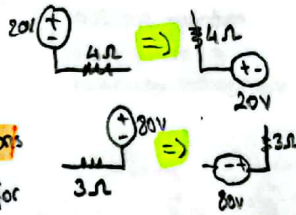
$$-i_2 + \frac{V_3 - V_2}{30} + \frac{V_3 - V_1}{20} = 0$$

$$i_2 = 3.6$$



Write the node-voltage equations and the constraint equation for the following circuit.

V_1 'e direkt 20V
 V_3 'e direkt 80V
denilemez. aşağıdaki dirençleri hiçe sayalım oluruz.



Constraint equation: $i_x = \frac{V_2 - 0}{25}$

KCL for node-1

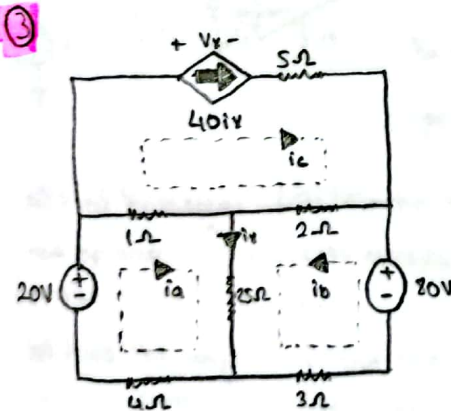
$$\frac{V_1 - 20}{4} + \frac{V_1 - V_2}{1} + 40i_x = 0 \quad 25V_1 + 12V_2 = 100$$

KCL for node-2

$$\frac{V_2 - V_1}{1} + \frac{V_2}{25} + \frac{V_2 - V_3}{2} = 0 \quad -50V_1 + 77V_2 - 25V_3 = 0$$

KCL for node-3

$$\frac{V_3 - V_2}{2} + \frac{V_3 - 80}{3} - 40i_x = 0 \quad -63V_2 + 25V_3 = 800$$



Write the mesh-current equations and the constraint equation for the following circuit.

constraint equation: $i_x = i_a + i_b$, from special case of mesh-current method $\Rightarrow i_c = 40i_x$

KVL for loop-1

$$-20 + (i_a - i_c) + 25(i_a + i_b) + 4i_a = 0 \quad 30i_a + 25i_b - i_c = 20$$

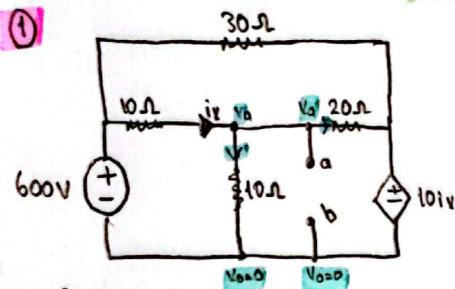
KVL for loop-2

$$2(i_b + i_c) + 25(i_a + i_b) + 3i_b - 80 = 0 \quad 25i_a + 30i_b + 2i_c = 80$$

KVL for loop-3

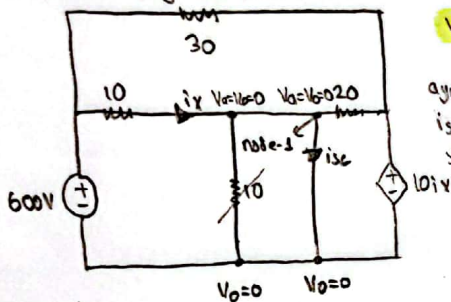
$$(i_c - i_a) + V_x + 5i_c + 2(i_c + i_b) = 0$$

CIRCUIT 2021 FINAL



Find the Thevenin equivalent seen from the terminals "a" and "b" of the following circuit.

now we apply isc from a to b



Constraint equation: $i_x = \frac{600 - V_a}{10}$

$V_a \rightarrow$ reference node

$V_a = V_{th}$

KCL for node-a

$\frac{V_a - 0}{10} + \frac{V_a - 10}{20} = \frac{600 - V_a}{10}$

$V_a = 300V = V_{th}$

$i_x = \frac{600 - V_a}{10}$

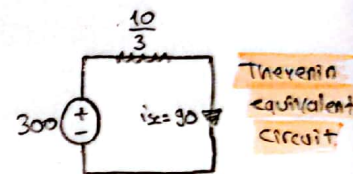
$i_x = 30$

KCL for Node-b

$i_{sc} + \frac{V_a - 10}{20} = i_x$

$i_{sc} = 30$

$R_{th} = \frac{V_{th}}{i_{sc}} = \frac{10}{3}$



Thevenin equivalent circuit

$V_a = V_b = 0$

1 / aynı nodlar isc'nin olduğu yerden kayarız

üstünü aşağıya 10Ω'lık yerden atım çarparız ve alabiliriz Voltage yok

Op-amp is ideal: $i_p = i_n = 0$

linear region: $-15 \leq V_o \leq 15$: $V_p = V_n$
: $V_p = 0, V_n = 0$

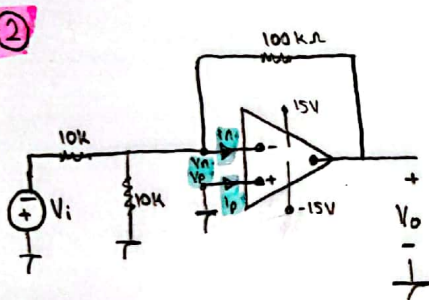
KCL for node-1

$\frac{V_n - (-\frac{V_i}{2})}{5} + \frac{V_n - V_o}{100} - i_n = 0$

$10V_i = V_o$

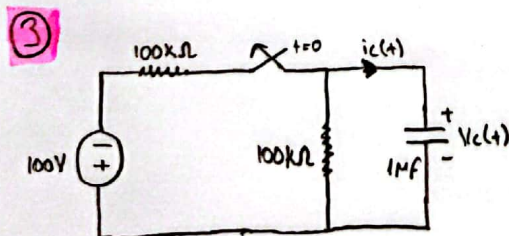
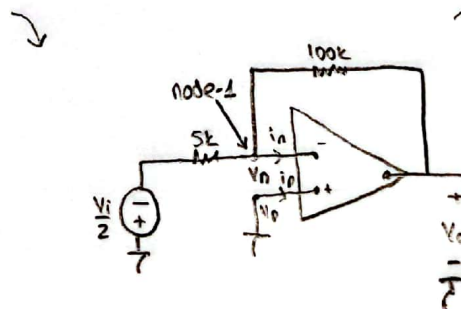
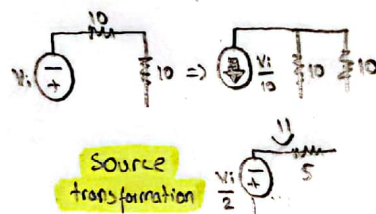
$-15 \leq V_o \leq 15$

$-1.5 \leq V_i \leq 1.5$



a) Find V_o in terms of V_i (Assume that the op-amp is ideal and the op-amp works in the linear region)

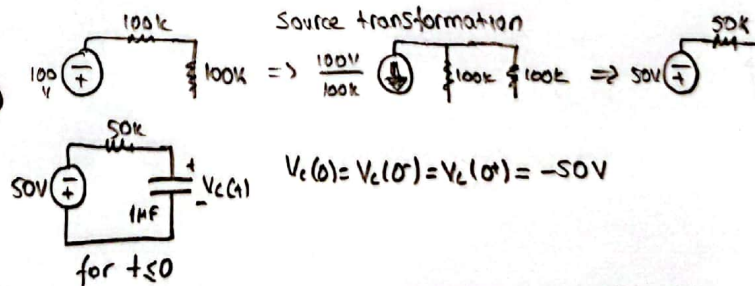
b) Find the range of V_i that the op-amp operates in its linear region.



The switch has been closed for a long time before opening at time $t=0$.

a) $V_c(t) = ?$ for $t \geq 0$

b) $i_c(t) = ?$ for $t \geq 0$

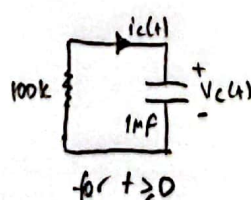


$V_c(0) = V_c(0^-) = V_c(0^+) = -50V$

$\tau = R \cdot C = 0.1 \text{ sec}$

$V_c(t) = V_c(0) e^{-\frac{t}{\tau}} = -50 e^{-10t}$

$i_c(t) = C \cdot \frac{dV_c(t)}{dt} = 1 \cdot 10^{-6} \cdot (-500 e^{-10t}) = \frac{-500 e^{-10t}}{2 \cdot 10^3}$



4 A ZAR

isa can Gabuk

AYRILA TESEK KÜALER

zekeriya TUFENGİ